

Rmk:

①  $\sum a_n$  (absolute) converge  $\nRightarrow \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = p < 1$

Example 1:  $a_n = \frac{1}{n^2}$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 1$$

$\nRightarrow \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = p < 1$

Example 2:  $a_n = \frac{\sin n}{n^2}$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{\sin n}{\sin(n+1)} \cdot \frac{(n+1)^2}{n^2} \text{ 不存在!}$$

课本作业

Sec 10.5

$$53. a_1 = \frac{1}{3}, \quad a_{n+1} = \sqrt[n]{a_n}$$

Sol:  $a_1 = \frac{1}{3}, \quad a_2 = \sqrt[2]{\frac{1}{3}}, \quad a_3 = \sqrt[3]{\sqrt[2]{\frac{1}{3}}}$

$$\dots a_n = \sqrt[n!]{\frac{1}{3}} = \frac{1}{3^{\frac{1}{n!}}}$$

$$\frac{1}{3^{\frac{1}{n}}} \leq a_n \leq 1$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ 1 & 1 & 1 \end{array} \quad n \rightarrow \infty$$

# 2024 Calculus II Supplement Homework (3)

1. Which of the following series converge absolutely, which converge conditionally, and which diverge? Give your reasons.

(a)  $\sum_{n=1}^{\infty} \frac{(-1)^n n}{2n^2+n+1}$ . (b)  $\sum_{n=2}^{\infty} \frac{(-1)^n}{n^p (\ln n)^q}$  ( $p > 0, q > 0$ ). (c)  $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^{p+\frac{1}{n}}}$  ( $p > 0$ ).

2. Exercise 50, (1),(2),(3),(7) in page 29 from Calculus Supplement Homework.

3 Let  $\sum_{n=0}^{\infty} \frac{a_n}{n!} x^n$  and  $\sum_{n=0}^{\infty} \frac{b_n}{n!} x^n$  be power series. Find  $\left( \sum_{n=0}^{\infty} \frac{a_n}{n!} x^n \right) \left( \sum_{n=0}^{\infty} \frac{b_n}{n!} x^n \right)$  and find all  $x$  such that the equality holds.

4. Suppose that a power series  $\sum_{n=0}^{\infty} c_n x^n$  with its the radius of convergence  $R > 0$  and its sum function  $f(x)$  is a odd (even, resp.) function. Prove that  $c_n = 0$  for every even (odd, resp.) number  $n$ .

Q1.

(a) 条件收敛, 因为  $\lim_{n \rightarrow \infty} \frac{|a_n|}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{n^2}{2n^2+n+1} = \frac{1}{2}$   
 并且  $\frac{n}{2n^2+n+1}$  单调递减,  $\lim_{n \rightarrow \infty} \frac{n}{2n^2+n+1} = 0$ .

17

(b)  $p > 1$  绝对收敛  
 $p=1, q > 1$  绝对收敛  
 其余情况 条件收敛

17

(C) let  $a_n = \frac{(-1)^n}{n^{p+\frac{1}{n}}}$ ,  $\lim_{n \rightarrow \infty} \frac{|a_n|}{\frac{1}{n^p}} = \lim_{n \rightarrow \infty} \frac{1}{n^{\frac{1}{n}}} = 1$

$\therefore \sum a_n$  absolutely converge if  $p > 1$

$\sum p_n$  diverge if  $0 < p \leq 1$

but when  $0 < p \leq 1$ ,

$$\frac{1}{n^{p+\frac{1}{n}}} = e^{-(p+\frac{1}{n})\ln n}$$

let  $f(x) = -(p + \frac{1}{x}) \ln x$

$$f'(x) = -\left[\left(p + \frac{1}{x}\right)\frac{1}{x} - \frac{\ln x}{x^2}\right] = \frac{\ln x - 1 - px}{x^2} < 0 \text{ for } x \gg 0.$$

$\therefore \frac{1}{n^{p+\frac{1}{n}}}$  monotonic decrease for  $n \gg 0$ .

$$\therefore \lim_{n \rightarrow \infty} \frac{1}{n^{p+\frac{1}{n}}} = 0$$

$\therefore$  By alternating series test,  $\sum a_n$  converges and hence conditionally converges.

□

Q2

50. Find the radius and interval of convergence of the following series.

$$(1) \sum_{n=1}^{\infty} \left( \frac{n}{n+1} \right)^{n^2} x^n;$$

$$(2) \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!};$$

$$(3) \sum_{n=1}^{\infty} \left( \sum_{k=1}^n \frac{1}{k} \right) (x-2)^n;$$

$$(4) \sum_{n=1}^{\infty} \frac{3^n}{2^n n!} (x-1)^n;$$

$$(5) \sum_{n=1}^{\infty} \frac{\ln^2 n}{n^n} x^{n^2};$$

$$(6) \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n}}{n 2^n};$$

$$(7) \sum_{n=1}^{\infty} \left( \frac{a^n}{n} + \frac{b^n}{n^2} \right) x^n \quad (a > 0, b > 0).$$

$$(1) \quad \frac{1}{R} = \lim_{n \rightarrow \infty} \sqrt[n]{\left( \frac{n}{n+1} \right)^{n^2}} = \lim_{n \rightarrow \infty} \left( \frac{n}{n+1} \right)^n = e^{-1}$$

$$\therefore R = e$$

$$\text{Interval: } (-e, e)$$

$$\text{Let } a_n = \left( \frac{n}{n+1} \right)^{n^2} e^n = \left( \frac{e}{\left( \frac{n+1}{n} \right)^n} \right)^n$$

where  $\left( \frac{n+1}{n} \right)^n \nearrow e$  by basic inequality:

$$\sqrt[n]{x_1 \cdots x_n} \leq \frac{x_1 + \cdots + x_n}{n}, \text{ then}$$

$$\left( \frac{n+1}{n} \right)^n \cdot 1 \leq \left( \frac{n \left( \frac{n+1}{n} \right) + 1}{n+1} \right)^{n+1} = \left( \frac{n+2}{n+1} \right)^{n+1}$$

$$\therefore \frac{e}{\left( \frac{n+1}{n} \right)^n} > 1 \quad \text{for each } n \in \mathbb{N}.$$

$$\text{then } \lim_{n \rightarrow \infty} a_n > \lim_{n \rightarrow \infty} 1^n = 1 \neq 0$$

$$\therefore \begin{cases} \sum a_n & \text{发散} \\ \sum (-1)^n a_n & \text{发散} \end{cases}$$

□

$$\begin{aligned}
 (2) \quad & 1 > \lim_{n \rightarrow \infty} \frac{x^{2n+2} (2n)!}{(2n+2)! x^{2n}} \\
 & = \lim_{n \rightarrow \infty} \frac{x^2}{(2n+2)(2n+1)} \\
 & = 0 \\
 \therefore R & = +\infty
 \end{aligned}$$

Interval :  $(-\infty, +\infty)$ .

$$\begin{aligned}
 (3) \quad & \lim_{n \rightarrow \infty} \left(1 + \frac{1}{2} + \dots + \frac{1}{n}\right)^{\frac{1}{n}} = 1 \\
 \therefore R & = 1
 \end{aligned}$$

Interval  $(1, 3)$

$$(7) \quad C_n = \frac{a^n}{n} + \frac{b^n}{n^2} = \frac{na^n + b^n}{n^2}$$

$$\lim_{n \rightarrow \infty} \frac{C_{n+1}}{C_n} = \lim_{n \rightarrow \infty} \frac{(n+1)a^{n+1} + b^{n+1}}{na^n + b^n} \cdot \frac{n^2}{(n+1)^2}$$

① assume  $a \geq b$

$$\text{原式} = \lim_{n \rightarrow \infty} \frac{\left(1 + \frac{1}{n}\right)a + \left(\frac{b}{a}\right)^n \cdot b \cdot \frac{1}{n}}{1 + \left(\frac{b}{a}\right)^n \cdot \frac{1}{n}} = a \quad R = \frac{1}{a}, \text{ interval } \left[-\frac{1}{a}, \frac{1}{a}\right)$$

② assume  $b > a$

$$\text{原式} = \lim_{n \rightarrow \infty} \frac{\left(1 + \frac{a}{b}\right)^n a + b}{n\left(\frac{a}{b}\right)^n + 1} = b \quad R = \frac{1}{b}, \text{ interval } \left[-\frac{1}{b}, \frac{1}{b}\right]$$

Reason for (3)

$$b_n = \sum_{k=1}^n \frac{1}{k} \quad \text{Then} \quad \sum_{n=1}^{\infty} b_n = \infty$$

$$C_n = \sum_{k=1}^n \frac{1}{k} \cdot (-1)^n$$

$$\therefore C_{2k-1} + C_{2k} = \frac{1}{2k}$$

$$\sum_{n=1}^{\infty} C_n = \sum_{k=1}^{\infty} C_{2k-1} + C_{2k}$$

$$= \sum_{k=1}^{\infty} \frac{1}{2k}$$

$$= \infty$$

Q3.

Sol.:  $\left( \sum_{n=0}^{\infty} \frac{(ax)^n}{n!} \right) \left( \sum_{n=0}^{\infty} \frac{(bx)^n}{n!} \right) = e^{ax} \cdot e^{bx} = e^{(a+b)x}$

which is hold for all  $x \in (-\infty, +\infty)$ .

□

Q4.

$$f(x) = \sum_{n=0}^{\infty} c_n x^n \text{ is odd,}$$

$$f(-x) = \sum_{n=0}^{\infty} (-1)^n c_n x^n$$

$$0 = f(x) + f(-x) = \sum_{n=0}^{\infty} (1 + (-1)^n) c_n x^n = \sum_{n \text{ even}} 2c_n x^n$$

$\therefore c_n = 0$  for  $n$  even.

□